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FFIEC

Retail Payment Systems

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IT EXAMINATION

HANDBOOK

Table of Contents

Introduction	1
Retail Payment Systems Overview	2
Payment Instruments, Clearing, and Settlement	4
Check-Based Payments	6
Remotely Created Checks	8
Electronically Created Payment Orders	10
Remote Deposit Capture	10
Check Clearing Houses	11
The Automated Clearing House (ACH)	13
The ACH Network	13
NACHA Rule and Product Changes	16
Card-Based Electronic Payments	17
General Purpose Credit Cards	18
Co-Branded/Affinity Credit Cards	19
Debit and ATM Cards	21
EFT/POS Networks	22
Prepaid (Stored Value) Cards	24
Payroll Cards	25
General Spending Reloadable Cards	27
Online Person-to-person (P2P), Account-to-Account (A2A) Payments and Electronic Cash	30
Emerging Retail Payment Technologies	32
Contactless Payment Cards, Proximity Payments and Other Devices	33
Biometrics for Payment Initiation and Authentication	33
Emerging Network Technologies	33
Retail Payment Systems Risk Management	35
Payment System Risk (PSR) Policy	36
Strategic Risk	37

Reputation Risk	38
Credit Risk	38
Liquidity Risk	40
Legal (Compliance) Risk	40
Operational Risk	42
Audit	44
Information Security	45
Business Continuity Planning	47
Vendor and Third-Party Management	48
Retail Payment Instrument Specific Risk Management Controls	49
Checks	49
ACH	50
Third-Party ACH Processing	52
Credit Cards	53
Debit/ATM Cards	54
Card/PIN Issuance	54
Merchant Acquiring	55
EFT/POS and Credit Card Networks	60
Appendix A: Examination Procedures	A-1
Appendix B: Glossary	B-1
Appendix C: Schematic of Retail Payments Access Channels & Payments Method	C-1
Appendix D: Laws, Regulations, and Guidance	D-1
Appendix E: Mobile Financial Services	E-1

Introduction

The FFIEC IT Examination Handbook (IT Handbook), "Retail Payment Systems Booklet" (booklet), provides guidance to examiners, financial institutions, and technology service providers (TSPs) ^[1] on identifying and controlling risks associated with retail payment systems and related banking activities. ^[2]

Financial institutions accept, collect, and process a variety of payment instruments and participate in clearing and settlement systems. In some cases, financial institutions perform all of these tasks. However, independent third parties are increasingly involved in this process, introducing new risks that affect the security of financial institutions. Financial institutions, acting either in consortiums or independently, remain the core providers to businesses and consumers for most retail payment instruments and services. Federal government-affiliated providers and operators, such as the Federal Reserve Banks (Reserve Banks), also compete with numerous financial institutions and private sector firms in providing various services in support of retail payments.

Recently, a number of new payment instruments have emerged that are largely or wholly electronic. Electronic payment systems offer efficiency gains by allowing for rapid and convenient transmission of payment information among system participants. However, the emergence of a new payment mechanism can also enable the rapid propagation of fraud, money laundering, and operational disruption if data is compromised. Another trend associated with emerging payments is the increased participation of nonbank third parties in retail payment systems and a lengthened transaction chain, which may increase risk in payment processes. Management of retail payments risk is increasingly difficult and requires diligent oversight of third-party service providers.

Much of the guidance in this booklet, involving traditional retail payment systems, has not been revised significantly because of the maturity of these systems in the product life cycle. Mature payment systems are better understood, whereas emerging payment systems require a closer look to better understand the risks and associated controls. New guidance is offered for remotely created checks (RCCs), electronically created payment orders, automated clearing house (ACH) transactions, The Check Clearing for the 21st Century Act (Check 21), ^[3] and Merchant Card Processing due to recent developments in these areas. Also, this booklet includes a new section that covers some emerging technologies in retail payment systems. Additional emphasis is placed on the need for improved operational, credit, legal, and compliance risk processes for retail payment products, especially for the deployment of remote and Internet-based check and ACH capture systems.

Examination guidance for Retail Payment Systems is provided in three sections, followed by examination procedures, a glossary, and references:

- Retail Payment Systems Overview-The first section of the booklet presents an overview of retail payment systems, grouping retail payment instruments in various categories, including: checks, card-based electronic payments, and other electronic payments, such as person-to-person (P2P), electronic benefits transfer (EBT), and ACH.
- Payment Instruments, Clearing, and Settlement-The second section of the booklet

describes the retail payment system instruments typically offered by financial institutions and the roles of various payment system participants, including third parties. Diagrams showing the typical payment flows and clearing and settlement arrangements for each of the retail payment instruments described are also included.^[4]

- **Retail Payment Systems Risk Management**-The third section describes the risks associated with various retail payment systems and instruments, using the regulatory risk categories: reputation, strategic, credit, liquidity, settlement, legal/compliance, and operational/transaction risk. This section also presents the risk management practices financial institutions should implement in order to mitigate the risks described, and it concludes with specific controls appropriate to a number of retail payment instruments. Management action summaries for selected risks and functions are also included in this section, providing a snapshot of the risks and risk management practices described in the text.

This booklet includes a number of references to other IT Handbook booklets, including "Information Security," "Business Continuity Planning," "Audit," "Outsourcing Technology Services," "Electronic Banking," and "Wholesale Payment Systems." Also, there are references to FFIEC guidance for Bank Secrecy Act examinations that are relevant to retail payment systems and for Check 21. In addition to describing the IT risks and controls, the booklet also discusses certain credit and liquidity risks that may also be present when providing retail payment services. A full review of a particular financial institution's retail payment system environment will require an interdisciplinary team of examiners with experience in operational, credit, liquidity, and compliance risks.

Examiners should use the examination procedures for evaluating the risks and risk management practices at financial institutions offering retail payment system products and services. These procedures address services and products of varied complexity; therefore, examiners should adjust the procedures, as appropriate, for the scope of the examination and the risk profile of the institution. The procedures may be used independently or in combination with procedures from other IT Handbook booklets and agency-specific handbooks and guidance documents.

Retail Payment Systems Overview

Retail payments usually involve transactions between two consumers, between consumers and businesses, or between two businesses. Wholesale payments are typically made between businesses. Although there is no definitive division between retail and wholesale payments, retail payment systems generally have higher transaction volumes and lower average dollar values than wholesale payment systems. This section provides background information on payments typically classified as retail payments. The following are examples of typical retail payments. These retail payments may involve the use of various retail payment instruments or access devices (e.g., checks, ACH, card, phones, etc.).

Purchase of Goods and Services-Purchase of goods and services can occur at the point-of-sale (POS) (e.g., in person at a merchant location, through the Internet, or by telephone). These payments include attended POS payment transactions for goods or services, such as with traditional retailers, and unattended payment transactions, as with

vending machines. Increasingly, traditional retailers such as grocers and home improvement stores are using unattended payment systems at the POS as well. As technology advances, the consumer can purchase goods and services remotely without physical presence at the POS, such as via the Internet or a telephone/mobile phone. Payment instruments for retail purchases of goods and services have expanded beyond traditional vehicles (i.e., cash, checks, and credit and debit cards) to prepaid cards, contactless debit and credit cards, and other contactless devices such as key fobs, mobile phones. In addition, merchants may convert checks to electronic form at the POS, and use the ACH system for clearing and settlement.

Bill Payment-Consumers may elect to pay (or provide payment instructions for) recurring or nonrecurring bills and invoices via electronic bill payment. A particular biller's periodic recurring invoices can be electronically paid individually or set up to be paid automatically to a payment schedule. In recent years, there has been a growing trend toward payment of recurring and nonrecurring bills using Internet-based bill payment services.

P2P Payments-The vast majority of consumer-to-consumer payments are conducted with checks and cash, with some transactions using electronic P2P payment systems. The expansion of systems that permit customers to conduct P2P payments is anticipated through account-to-account (A2A) transfers, which use either the ACH or Automated Teller Machine (ATM) networks for movement of funds.

A2A Payments-With A2A payments, the consumer moves funds from his or her account at a financial institution to the account of another individual or business at the same or a different financial institution. The emerging use of the ATM networks for movement of funds may allow same day availability of funds at a cost far less than traditional wire transfer systems.

Cash Withdrawals and Advances-Consumers use retail payment instruments to obtain cash from merchants or ATMs. For example, consumers can use a credit card to obtain a cash advance through an ATM or an ATM or debit card to withdraw cash from an existing account. Consumers can also use personal identification number (PIN)-based debit cards to withdraw cash at an ATM or receive cash back at some POS locations.

Retail payment systems continue to evolve with advances in technology. These advances enable financial institutions to develop new products and services, lower the barriers to business entry for smaller institutions, and exploit economies of scale.

Recent changes in payments technology have influenced three important trends in retail payments. First, as firms seek economies of scale, the banking industry has witnessed the rapid consolidation of retail payment service providers, credit issuers, merchant acquirers, processing companies, and check processors. As a result, some small and mid-sized financial institutions have exited the business and outsourced certain functions of the retail payments process to larger financial and non-financial institutions. Nonbanks, in particular, are assuming more roles in retail payment systems such as the clearing and settlement payment functions and the issuance and processing of electronic payment cards and other devices.

The second trend is the shift from paper to electronic payments as technology has converged with the change in consumers' and merchants' preferences for convenient and low cost payment alternatives. The most significant growth is seen in debit and prepaid cards (stored value cards), followed by the increased use of Internet services like online banking and bill pay. The volume of checks and cash payments continues to decrease, with cash usage declining at a much slower rate. The emergence of new

electronic payment vehicles in the U.S. is anticipated as they are adopted in the global market.

Use of automated bill pay is a third important trend. Although consumers traditionally used checks for a large portion of bill payments in the U.S., direct bill payment through the ACH system are increasingly popular. More recently, retail firms have used check-to-ACH conversion processes to allow electronic settlement, thereby reducing the number of checks that flow through the payment system.

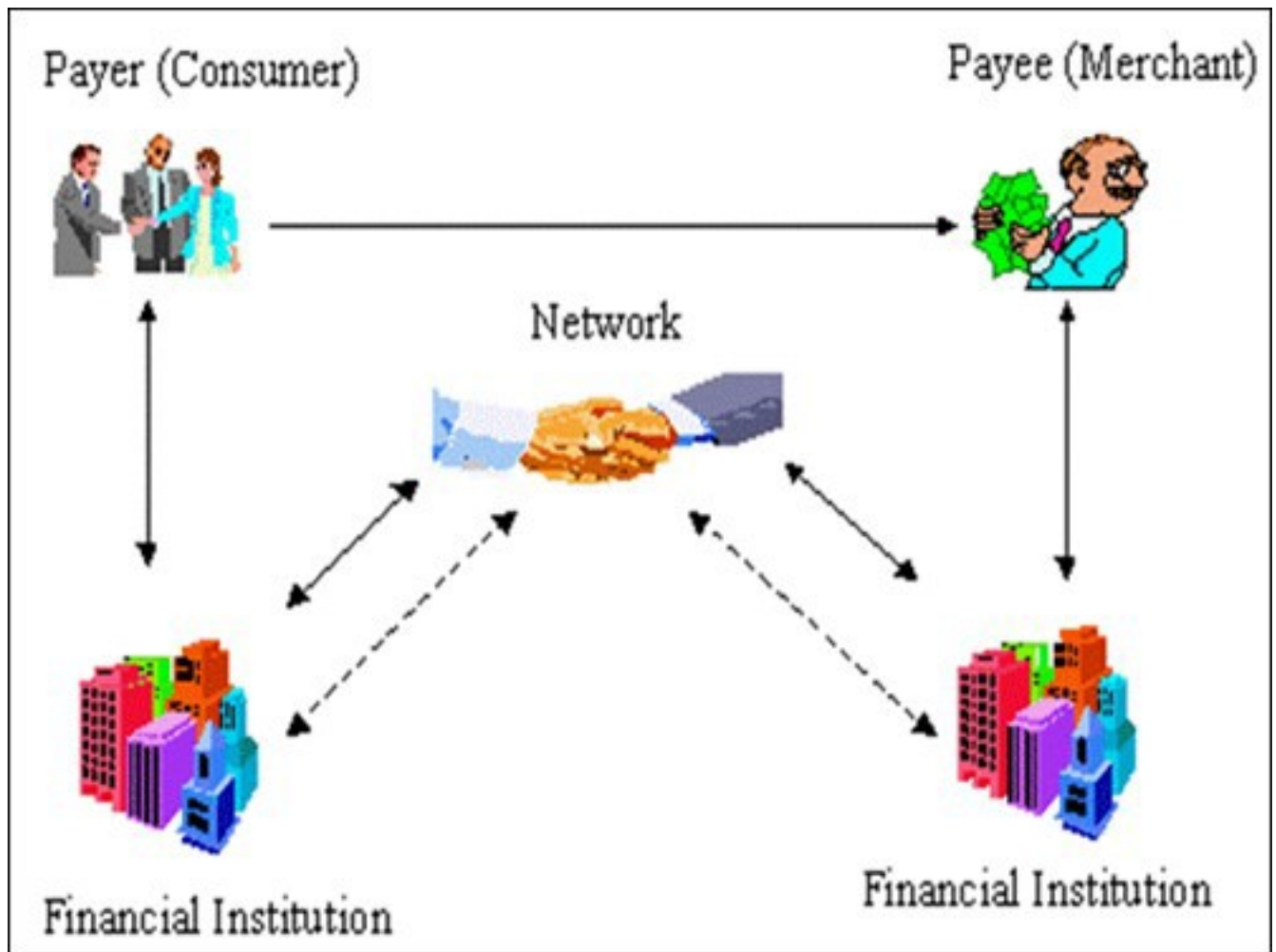
International retail payments are relatively new in the ACH industry and are largely driven by businesses and consumers seeking cost reductions for funds transfers across borders. Several financial institutions maintain their own proprietary systems, and more recently the Reserve Banks began offering FedACH International Services. FedACH International provides a means of transmitting funds between the U.S. and other countries using NACHA - The Electronic Payments Association (NACHA) rules. ^[5]

Beginning September 18, 2009, a new Standard Entry Class (SEC) code became effective that is expected to facilitate compliance due diligence with the use of the ACH system for international payments. The International ACH Transaction SEC code (IAT) will enable financial institutions to identify international ACH payments and perform the due diligence required by the U.S. Office of Foreign Assets Control.

Consumer and merchant acceptance of all the technological changes has been vital to the success of emerging retail payment systems and products. Consumers have shown willingness to accept new retail payment technologies more quickly because of the convenience afforded by these new services.

Payment Instruments, Clearing, and Settlement

This section provides an overview of the various payment instruments and clearing and settlement processes used for different retail payment systems. Although the diagrams reflect the general flow of transactions and participants, in many cases, other third parties may facilitate one or more processing functions.



Legend: Solid lines represent the flow of information and dashed lines represent the flow of funds.

Figure 1: Four-Corner Payments Model

Figure 1 displays the clearing and settlement process for retail payments using a standard four-corner payments model. While the flow of information and funds is different for each payment instrument, there is a common set of participants for retail payments. The initiator of the payment, typically a consumer, is located in the upper left-hand corner of the diagram. The recipient of the payment, typically a merchant, is in the upper right-hand corner of the diagram. The lower two corners of the model represent the relationship of the consumer and merchant with their financial institutions. The payments networks or clearing house organizations that route the transactions between financial institutions are in the middle of the chart. In subsequent model figures, solid lines represent the flow of information, and dashed lines represent the flow of funds. This generic figure can be applied to all retail payments.

More financial institutions are engaging third-party service providers to act on their behalf rather than keeping all payment functions in-house. In some instances, such as in check

clearing, a financial institution may exchange check items directly with another financial institution without using an intermediary.

There are a variety of retail payment clearing and settlement systems. These include; check clearing systems, ACH networks, ATM networks, and bankcard networks. Check clearing systems can be paper-based or electronic. Check 21 is facilitating the expanded use of electronic imaging technologies in check processing, enabling the banking industry to improve the efficiency and cost-effectiveness of check processing operations.

ACH payments also have grown significantly as consumers are using more direct bill payments through the ACH. More recently, retail firms have employed check-to-ACH conversion processes to obtain the efficiencies of electronic processing, reducing the number of checks that flow through the payment system.

Internet-based bill payment systems are transaction origination platforms that allow customers to initiate bill payments through existing payment systems. Depending on the bill payment software implemented, the payment transaction may be processed through ATM, ACH, or check systems. ^[6] The following sections describe these systems in more detail.

Debit and credit cards, particularly signature and PIN debit, have driven much of the growth in electronic payments. The recent introduction of contactless payment cards is expected to contribute to the increase of merchant acceptance and financial institution issuance of cards and investment in contactless payment infrastructure.

Retail payments often move through multiple channels, which results in data being processed and stored on multiple systems that are typically outside of the direct control of the customer's financial institution. There are two primary challenges for financial institutions in managing these complex payment systems. First, the lack of interoperability ^[7] that often characterizes these systems and the associated lack of optimal data protocols may result in data integrity issues. Second, the complexity of systems increases the difficulty of the management of data security and system availability.

Check-Based Payments

Checks are the traditional method that consumers can use to access their accounts. A check contains the names of the payer and the payee, the payer's account number, amount of the check, and the name and routing number of the paying financial institution. The magnetic ink character recognition (MICR) line at the bottom of the check enables high-speed reader/sorter equipment to process checks. Before financial institutions process checks, they encode the amount of the check in magnetic ink at the bottom of the check. Check formats are governed by standards developed by the Accredited Standards Committee (ASC) on Financial Services, X9B Committee, which works under procedures sanctioned by the American National Standards Institute (ANSI). ^[8]

Check processing has undergone a transformation during the past five years; a trend that is expected to continue for the next several years. Until recently, consumers in the United States used checks more often than any other retail payment instrument other than cash. However, in an increasing number of payment situations, checks are no longer the most convenient payment instruments for consumers, or the most cost-

effective payment method for financial institutions and merchants. Checks comprise a decreasing percentage of the total noncash payment volume in the United States. Many consumers use checks merely for person-to-person transactions that are not conducive to electronic payments, and have shifted to electronic payments for POS transactions and bill payment. In addition, a significant volume of checks are converted to ACH debits at POS and at lock-box operations.

Legal developments have affected the processing of checks as well. Check 21, which became effective on October 28, 2004, has succeeded in reducing check processing times as well as the float period previously associated with physical processing. By authorizing the use of a new negotiable instrument called a substitute check, Check 21 facilitates the broader use of electronic check processing.

A properly-prepared substitute check is the legal equivalent of the original check and includes all the information contained on the original check. The law does not require financial institutions to accept checks in electronic form, nor does it require financial institutions to use the new authority granted by the act to create substitute checks. The law permits financial institutions to truncate ^[9] original checks, process the check information electronically, and deliver substitute checks to financial institutions that wish to receive paper checks in lieu of electronic alternatives.

For many financial institutions, implementing a Check 21 strategy involves a significant investment in new hardware and software as well as the reengineering of check processing routines. Consequently, financial institutions should deploy Check 21 with appropriate risk management, including strategic planning, project management, and vendor management. Check 21 requires the bank ^[10] that creates a substitute check, the reconverting bank, to warrant that there will not be duplicate presentments of the check (or copy or representation thereof) and that the substitute check is an accurate representation of the original check as of the time the original check was truncated. Such substitute checks must meet specific requirements to be treated as a legal equivalent, and the bank that creates a substitute check must indemnify other parties for losses that result from their receipt of a substitute check instead of the original check.

Financial institutions implementing a Check 21 strategy must consider new processes for imaging checks, transferring files of imaged checks, and archiving and retrieving imaged checks. For example, a number of financial institutions are implementing remote check capture systems in their branches and processing centers as a means of significantly reducing check transit costs. Some financial institutions are providing selected customers with remote check capture devices. Examiners are encouraged to review the FFIEC's guidance for Risk Management of Remote Deposit Capture. ^[11]

Another important catalyst for the changes taking place in payment systems is electronic check conversion, a process in which information from a check is used to create an ACH debit. The conversion may occur at a retailer's POS, or at lock-box processing centers to which a consumer mails checks. Electronic check conversion is similar to, but separate from, the check substitution process authorized by Check 21. Instead of using the image of a paper check, as in the Check 21 process, the recipient uses the account and financial institution information contained on the consumer's check to create a new electronic payment through either the ACH or debit card networks. ^[12]

ACH electronic fund transfers between financial institutions are not considered check transactions; thus, they are not subject to laws governing check processing. Rather, they are governed by the rules of the ACH that processes the electronic fund transfer. ACH transactions to or from consumer accounts also are subject to the provisions of the Federal Reserve Board's Regulation E, Electronic Fund Transfers.